

AI-Driven Threat Detection in Cybersecurity: Investigating the Role of Machine Learning and Deep Learning in Enhancing Intrusion Detection Systems

Aven Pacha Pinto

Southeast Missouri State University

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Abstract— As digital technologies continue to grow, cybersecurity has become increasingly important for protecting networks and information systems from cyber threats. Traditional intrusion detection systems (IDS) often struggle to identify new attacks and may incorrectly classify normal network activity as a threat. To address these challenges, researchers have begun using Artificial Intelligence (AI), particularly Machine Learning (ML) and Deep Learning (DL), to improve how intrusion detection systems identify and respond to malicious activity. This paper examines how ML and DL techniques are being used to enhance intrusion detection in network and cloud environments. Through a review of six research studies, it explores the improvements offered by AI-driven systems, including higher detection accuracy, fewer false alarms, and the ability to recognize more complex attack patterns. The review also identifies several challenges, such as the need for large datasets, high computational requirements, limited interpretability, and vulnerabilities to adversarial attacks. Overall, the findings suggest that AI-driven approaches have the potential to improve intrusion detection, but further research is needed to make these systems more reliable, efficient, and secure in real-world environments.

Keywords—artificial intelligence, cybersecurity, deep learning, intrusion detection systems, machine learning

I. INTRODUCTION

In today's world, digital technology plays a central role in almost every part of modern life. Organizations, governments, and individuals depend heavily on information and communication technology (ICT) systems to store, process, and exchange information [1]. These systems include cloud computing, which allows data and applications to run on remote servers, and the Internet of Things (IoT), which connects millions of physical devices such as sensors, smart home appliances, and industrial equipment to the internet. Together, these technologies make our systems faster, more efficient, and more connected. However, this rapid expansion has also created new security risks. Each connected device, cloud service, or online application increases the number of possible entry points that hackers can exploit [2]. As a result, cybersecurity has become one of the most important areas in modern computing.

The volume of data moving through networks today is larger than ever before, and it continues to grow every year [1]. While this growth has many benefits, it also gives attackers more opportunities to hide malicious activity within normal data traffic. Modern cyber threats are not only more frequent but also far more sophisticated. For example, Distributed Denial of Service (DDoS) attacks flood a system with massive amounts of traffic until it crashes. Zero-day exploits take advantage of unknown software vulnerabilities before developers can fix them, and Advanced Persistent Threats (APTs) use stealthy, long-term strategies to infiltrate organizations and steal sensitive information [2]. The growing number and complexity of these attacks make it clear that traditional security defenses are no longer enough.

Intrusion Detection Systems (IDS) are one of the main defenses used to protect networks and cloud systems. Early IDSs used signature-based detection, which means they looked for patterns that matched known attacks [2]. These systems were effective at identifying known threats but were

less effective at detecting new or unfamiliar attacks. Later, anomaly-based IDSs were created to detect unusual behavior, but they often made mistakes by flagging normal activity as attacks. Because today's cloud systems handle large amounts of fast-moving data, these traditional methods are not always effective [3].

To solve these problems, researchers have started using Artificial Intelligence (AI), especially Machine Learning (ML) and Deep Learning (DL), to make IDSs smarter and more accurate. These systems have the ability to learn from data and recognize complicated patterns that older systems might miss [2]. AI-driven IDSs use models such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Long Short-Term Memory (LSTM) networks to improve how they identify and classify threats. These approaches have been shown to lower false alarms and adapt better to new types of attacks than signature-based systems [1].

This report focuses on how ML and DL are being used to improve intrusion detection in cloud and network environments. The main goal is to understand how AI-based systems perform compared to traditional IDS methods and how they handle current cybersecurity challenges.

The report aims to: (1) examine how ML and DL techniques enhance IDS and anomaly detection; (2) compare traditional signature-based systems with AI-driven approaches; and (3) explore current limitations, particularly adversarial attacks that target AI models and reduce their reliability.

The scope of this report includes network-based IDS (NIDS), host-based IDS (HIDS), and hybrid systems that integrate multiple learning architectures. This report reviews six research studies, including models such as DCNNBiLSTM [4], HDLNIDS [3], and DeepADV [5], along with two additional sources that provide background information on cybersecurity challenges and adversarial machine learning. Together, these sources highlight how hybrid deep-learning systems can improve intrusion detection while also identifying

important challenges, such as data imbalance, limited interpretability, and vulnerabilities to adversarial attacks.

However, AI-driven cybersecurity still has some important limitations. These models need large, high-quality datasets to perform well, and training deep learning models requires powerful computers [2], [3]. Attackers can also use Adversarial Machine Learning (AML) to mislead the model and cause incorrect outputs [6]. In addition, many advanced DL models lack interpretability and transparency, making it difficult for security analysts to understand or confirm the reasoning behind detection decisions [2].

This report first reviews research on machine learning and deep learning for intrusion detection, including reported advances in accuracy, scalability, and hybrid architectures. It then discusses dataset limitations, computational requirements, interpretability, adversarial machine learning, and research directions emerging from the reviewed studies.

II. LITERATURE REVIEW

A. Overview of AI-Driven Intrusion Detection Research

The studies reviewed demonstrate how ML and DL techniques can improve intrusion detection by identifying complex attack patterns and reducing false alarms. Several studies also show that hybrid models, which combine different learning techniques, can improve detection accuracy. However, the research highlights important challenges, including class imbalance in datasets, high computational requirements, and vulnerabilities to adversarial attacks.

B. Technical Advancements in AI-Driven Intrusion Detection

1) Hybrid Deep Learning Architectures

One major advancement is the use of hybrid DL architectures. These models combine different neural network types so they can learn different kinds of features from the data.

For example, the DCNNBiLSTM model by Hnamte and Hussain [4] combines Deep Convolutional Neural Networks (DCNNs) with Bidirectional Long Short-Term Memory (BiLSTM) networks. The DCNN captures local patterns in the data, while the BiLSTM learns how events unfold over time. With additional techniques such as attention mechanisms and batch normalization, the model achieved a reported accuracy of 100 percent on CICIDS2018 and 99.64 percent on Edge_IIoT.

Similarly, the HDLNIDS model by Qazi et al. [3] uses a hybrid CNN-RNN structure. The CNN extracts local features, while the RNN captures long-term relationships. The authors also used over-sampling and Gaussian noise layers to improve the model's flexibility and accuracy. HDLNIDS achieved an average accuracy of 98.90 percent using 10-fold cross-validation on the CICIDS-2018 dataset.

Sajid et al. [1] also support hybrid designs by combining ML models like XGBoost with DL components such as CNNs and LSTMs. The ML models help with feature selection, and the DL components help with pattern recognition and classification. These combinations are especially helpful for handling high-dimensional data and improving generalization.

In a different domain, Alladi et al. [5] introduced DeepADV for anomaly detection in Vehicular Ad-hoc Networks (VANETs). Their results show that hybrid models, such as CNN-LSTM, are effective even outside traditional network traffic environments.

2) Improvements in Accuracy and Lower False Positives

Several of the reviewed studies demonstrate how AI-based intrusion detection can improve detection accuracy and reduce false positives. These systems learn from large amounts of network data and can recognize subtle differences in behavior that traditional IDS methods might miss.

For example, Vinayakumar et al. [7] showed that Deep Neural Networks (DNNs) performed better than classical ML methods such as support vector machines and decision trees across several well-known datasets, including NSL-KDD, UNSW-NB15, and CICIDS 2017. Their results demonstrate that deeper and more flexible models can generalize better across different kinds of cyberattacks.

3) Scalability and Real-Time Processing

Another important improvement is the ability of some models to scale and support real-time analysis. As network data grows rapidly—especially in cloud environments—IDS models must process information quickly and efficiently.

Vinayakumar et al. [7] addressed this challenge through Scale-Hybrid-IDS-AlertNet (SHIA), a distributed deep learning framework. SHIA runs DNN components across multiple machines and supports both network-level and host-level monitoring. This approach makes it possible to analyze large amounts of data continuously, which is necessary in modern cloud and enterprise systems.

C. Key Challenges Identified in the Literature

Although the studies show strong progress, they also identify several challenges that limit real-world deployment of AI-driven IDS.

1) Dataset Limitations and Class Imbalance

AI models depend heavily on good-quality data. Many cybersecurity datasets suffer from class imbalance, meaning that normal traffic greatly outnumbers malicious traffic. For example, the CICIDS-2018 dataset used by Qazi et al. [3] contains relatively few attack samples. Oversampling techniques can help balance the data but may also increase the risk of overfitting.

Another issue is the continued use of older datasets, such as KDDCup 99. These datasets contain outdated attack patterns and repetitive samples, which limit the model's ability to detect modern threats. Vinayakumar et al. [7] stress the need for updated, realistic datasets.

2) High Computational Requirements

Many hybrid DL models require significant computing power and memory. This makes them difficult to deploy in environments without strong hardware support. Hnamte and Hussain [4], Sajid et al. [1], and Qazi et al. [3] all note that their models take a long time to train, and some require large amounts of energy for sustained processing. These factors can slow down real-time deployment, especially in cloud or edge environments.

3) Limited Interpretability

A major concern with deep learning is that the models often act as “black boxes.” This means analysts cannot easily understand how or why a model made a specific decision. Reddy et al. [2] note that this lack of transparency makes it harder to trust and verify AI-generated alerts. The need for Explainable AI (XAI) is therefore becoming an important direction for future work.

D. Vulnerabilities Introduced by AI-Driven Systems

1) Adversarial Machine Learning (AML)

AI models themselves can become new targets. Adversarial machine learning (AML) involves attacks that intentionally manipulate model inputs to cause wrong predictions.

Finlayson et al. [8] demonstrated this risk in the medical field. Using techniques such as Projected Gradient Descent (PGD) and adversarial patch attacks, they showed that even slight changes to an image can cause a DL model to make incorrect classifications. Although their work focuses on medical systems, the study demonstrates vulnerabilities that may also affect AI-driven intrusion detection systems.

Mohammed [6] describes how attackers can create adversarial examples to bypass AI-based IDS. These examples can be crafted even in black-box scenarios where the attacker does not know the exact model details. This makes AML an important and growing threat in cybersecurity.

2) Reliability and Trust Issues

AML attacks raise serious concerns about the reliability and trustworthiness of AI-based IDS. If attackers can fool these models, then organizations may not be able to depend on their predictions. Training-data poisoning, model manipulation, and generalization weaknesses all highlight the fragility of AI-driven approaches when facing skilled adversaries.

E. Research Gaps and Future Directions

1) Better Defenses Against Adversarial Attacks

Future research needs to focus on improving model robustness. Techniques such as adversarial training, input preprocessing, and stronger regularization may help models resist adversarial attacks.

2) Improved Detection of Zero-Day and Rare Attacks

Although hybrid models perform well overall, they still struggle with rare attacks, which appear infrequently in datasets. These low-frequency or zero-day attacks remain difficult to detect. Models like DCNNBiLSTM and HDLNIDS may benefit from new techniques that better capture unusual or emerging behaviors.

3) Optimizing Models for Real-Time and Large-Scale Use

More research is needed to reduce the computational load of DL-based IDS. Approaches such as model compression, efficient feature selection, and lightweight neural architectures could help bring these systems closer to real-time performance.

4) Expanding Explainable AI (XAI) for IDS

Because transparency and trust are essential in security environments, researchers should work on integrating Explainable AI into IDS models. This would help analysts

better understand why alerts occur and how the model makes its decisions.

F. Comparison of Reviewed Studies

TABLE I
SUMMARY OF SIX REVIEWED STUDIES

Study	Datasets, Methods, and Challenges
Sajid et al. [1]	CIC IDS 2017; UNSW NB15; NSL KDD; WSN DS. XGBoost-LSTM and CNN-LSTM; feature selection, class imbalance, false positives.
Finlayson et al. [8]	Medical imaging datasets. PGD and adversarial-patch attacks against ResNet-50-based classifiers; illustrates adversarial vulnerability.
Vinayakumar et al. [7]	KDDCup 99; NSL-KDD; UNSW-NB15; Kyoto; WSN-DS; CICIDS 2017; ADFA-LD/WD. Distributed DNN framework (SHIA).
Hnamte and Hussain [4]	CICIDS2018; Edge_IIoT. DCNN-BiLSTM with attention and batch normalization; high-dimensional threat detection.
Qazi et al. [3]	CICIDS-2018. Hybrid CNN-RNN with oversampling and Gaussian noise; class imbalance and low-frequency attacks.
Alladi et al. [5]	VeReMi extension dataset. DeepADV models including CNN-LSTM and CNN-BiLSTM; vehicular anomaly detection.

III. CONCLUSION

As cyber threats continue to evolve, traditional intrusion detection systems face increasing challenges in identifying new and complex attacks. This report examined how Machine Learning (ML) and Deep Learning (DL) can help address these challenges by improving the ability of intrusion detection systems to recognize unusual patterns and identify potential threats. The studies reviewed show that AI-driven approaches, particularly hybrid models that combine different learning techniques, can improve detection accuracy and reduce false alarms compared to traditional methods.

However, these improvements also introduce new challenges. AI-driven systems depend on large amounts of high-quality data, and some models require significant computing resources to train and operate effectively. Another important concern is that deep learning models can be difficult to interpret, making it harder for security analysts to understand why a particular activity was identified as a threat. Additionally, adversarial machine learning presents a new security risk because attackers can intentionally manipulate data to cause AI models to make incorrect predictions.

Although the research reviewed demonstrates the potential of AI in cybersecurity, further work is needed to address these limitations. Future research should focus on improving the detection of rare and zero-day attacks, reducing computational requirements, and developing stronger defenses against adversarial attacks. Improving the transparency of AI-driven systems is also important so that security analysts can better understand and verify their decisions.

Overall, ML and DL offer promising approaches for improving intrusion detection, but their effectiveness depends on more than detection accuracy alone. For these systems to be useful in real-world cybersecurity environments, they must also be reliable, efficient, and capable of adapting to new threats.

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